29 The ratio of ^{14}C : ^{12}C in living things is fixed when they are alive but decreases after death because of the radioactive decay of ^{14}C. ^{14}C has a half-life of 5730 years. A 10.0 g sample of wood cut from a live tree is measured to have an activity of 120 min^{-1}.

If a 1.0 g sample of an old wooden relic is found to have an activity of 8.05 min^{-1}, how long ago was this wooden relic cut from a live tree?

A 1470 years **B** 3300 years **C** 9950 years **D** 22300 years